

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 22)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1 Explain how each of the following building design features reduces energy use.

[6 marks]

Orientation
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Low surface area : volume ratio
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Earth sheltering
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0 1 . 2 Compare the embodied energy of concrete with alternative building materials.

[2 marks]

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0 1 . 3 Describe how a heat exchanger allows a building to be ventilated with minimal heat loss.

[2 marks]

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10

0 2

Table 1 compares two types of satellite used to monitor the Earth.

Table 1

Feature	Low Earth Orbit (polar) satellite	Geostationary satellite
Altitude	About 800 km	36 000 km above the equator
Orbit period	About 1.5 hours	24 hours (fixed position)
Coverage	Whole Earth surveyed in about 15 days	Same third of the Earth continuously
Detail	High resolution	Lower resolution

0 2**1**

Use Table 1 to explain why LEO satellites are used to monitor changes in ice cover and vegetation, whereas geostationary satellites are used to monitor weather.

[3 marks]

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0 2**2**

Give **three** climate-related factors, other than ice cover, that are measured by satellite sensors.

[3 marks]**1**

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0 2**3**

Explain how a computer climate model can be tested for reliability before its predictions are used.

[2 marks]

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0 2 . 4

Explain why representative data are essential for climate modelling.

[2 marks]

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10

0 3 . 1

Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

- A A waste is assumed to cause pollution until research shows it is unlikely to do so ☒
- B The polluter must pay for any damage caused ☐
- C Emissions should be as low as reasonably achievable ☐
- D Only pollutants known to be toxic need to be controlled ☐

0 3 . 2

Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

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0 3 . 3

Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

1

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2

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5

0 4 . A pumped-storage HEP station has an upper reservoir 500 m above its turbines. It can hold 7 million m^3 of water. The density of water is 1000 kg m^{-3} and $g = 9.8 \text{ N kg}^{-1}$.

0 4 . **1** Describe how pumped-storage HEP is used for peak shaving.

[3 marks]

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0 4 . **2** Calculate the maximum gravitational potential energy stored when the upper reservoir is full. Give your answer in standard form.

[3 marks]

Answer

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0 4 . **3** The station returns 75% of the electricity used for pumping. Explain why the process is still worthwhile despite this loss.

[2 marks]

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0 4 . **4** Give **two** locational requirements of a pumped-storage station.

[2 marks]

1

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10

0 5 A student compared the invertebrates in a stream above and below a sewage outfall using kick sampling.

0 5 . **1** Describe how kick sampling is carried out so that the samples from the two sites are comparable.

[3 marks]

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0 5 . **2** Give **three** limitations of kick sampling.

[3 marks]

1

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0 5 . **3** Explain how a Surber sampler improves on kick sampling.

[2 marks]

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0 5 . 4

Explain why the invertebrates found give a better indication of long-term sewage pollution than a single measurement of dissolved oxygen.

[2 marks]

10

0 6 . 1

Which of the following is the main reason that lead-based paint is a greater risk to children than to adults? Tick one box.

[1 mark]

A Children inhale more lead vapour

☐

B Children pick up dust / flakes and suck their fingers

☐

C Lead is more soluble in children's blood

☐

D Children spend more time near roads

☐

0 6 . 2

Explain how tri-butyl tin (TBT) in anti-fouling paint caused pollution even though only small amounts dissolved into the sea.

[3 marks]

0 6 . 3

Iron is not toxic, yet iron-rich drainage from mine spoil heaps kills fish. Explain why, and describe how it is controlled.

[3 marks]

06 . 4

Give **two** uses of lead that continue because they do not cause significant lead release into the environment.

[2 marks]

1

2

06 . 5

Explain why heavy metals are described as neurotoxins.

[1 mark]

10

07 . 1

Give **three** sources of unburnt hydrocarbons in urban air.

[3 marks]

1

2

3

07 . 2

Describe **two** methods used to control hydrocarbon emissions at petrol filling stations.

[2 marks]

1

2

0 7 . 3 Explain how a catalytic converter controls the emissions of carbon monoxide, hydrocarbons and NO_x.

[3 marks]

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0 7 . 4 Explain why carbon monoxide is toxic to humans.

[2 marks]

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10

0 8 . 1 Describe the process of anaerobic sludge digestion and give **two** benefits of the process.

[4 marks]

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08 . 2

Explain **two** problems of using sewage sludge as an agricultural fertiliser and how each can be reduced.

[4 marks]

08 . 3

Explain why it is more efficient to have separate collection systems for foul sewage and for surface drainage water.

[2 marks]**10**

09 . 1

Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

- A Absorbent mops
- B Steam washing
- C Inflatable booms
- D Polymerising agents

☐
☐
☐
☐

09 . 2

Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

09 . 3

Explain why deoxygenation is a more common problem in water than in air.

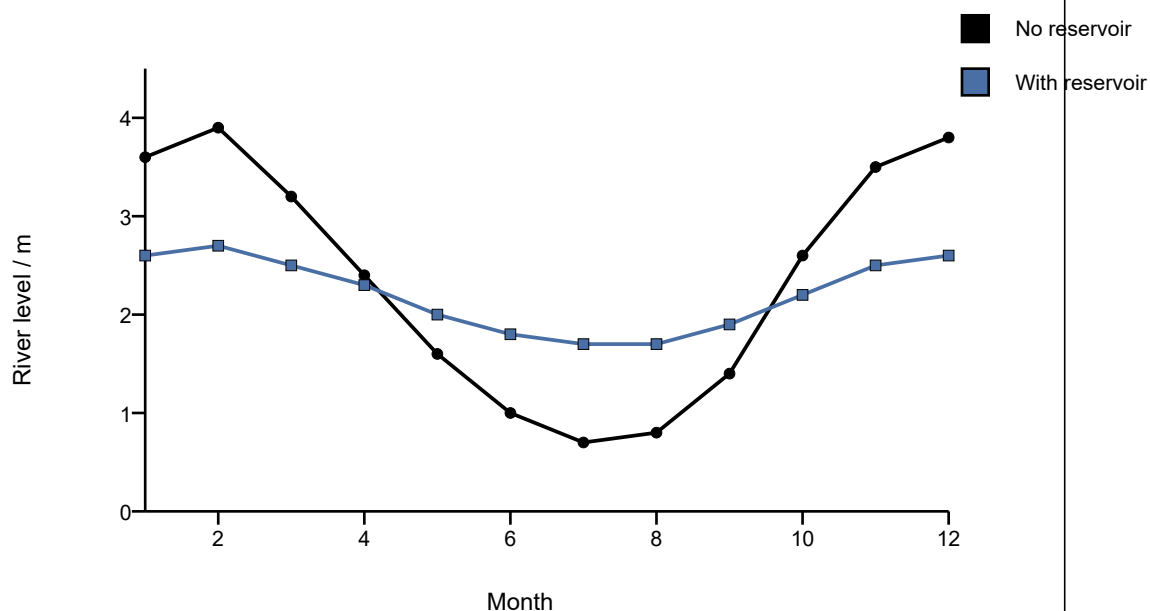
[2 marks]

5

10

Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



10 . 1

Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer

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1 0 . 2

Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

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1 0 . 3

Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

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1 0 . 4

Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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Extra space

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15

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

OR

1 1 . 2

Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

END OF QUESTIONS